

| Algorithmics | Student information     | Date                                                                                                                                          | Number of session |
|--------------|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
|              | UO: 307423              | 19/04                                                                                                                                         | 6                 |
|              | Surname: Miranda Martín |  Escuela de Ingeniería Informática<br>Universidad de Oviedo |                   |
|              | Name: Mariana Patricia  |                                                                                                                                               |                   |



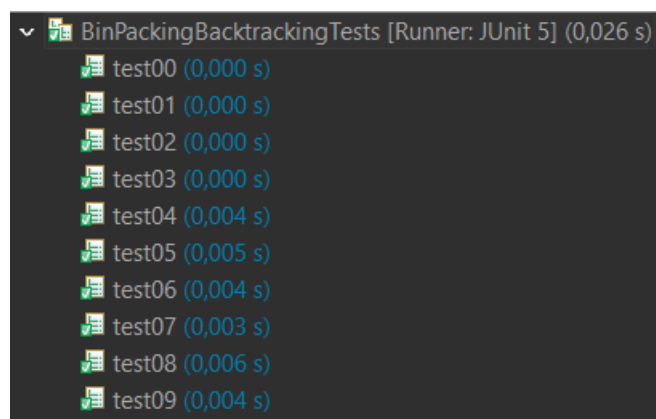
## Activity 1. [Bin Packing]

PART A: A class `BinPackingBacktracking.java` that after being executed by passing the file name as a program argument, it solves the problem as indicated.

A file with test cases named `BinPackingBacktrackingTests.java` is also provided. Only the path of the text files can be modified in that file. Test cases should be OK (although depending on the times obtained, not all of them may be able to be completed).

You can see `BinPackingBacktracking.java` implementation in [GitHub](#).

The tests are working as expected, as you can see below.



PART B: A class `BinPackingBacktrackingTimes.java` that will run all the test files to calculate the time needed to solve the problem. If the program takes too long for any of them, it can be stopped for that specific case.

You can see `BinPackingBacktrackingTimes.java` implementation in [GitHub](#).

| Algorithmics | Student information     | Date  | Number of session |
|--------------|-------------------------|-------|-------------------|
|              | UO: 307423              | 19/04 | 6                 |
|              | Surname: Miranda Martín |       |                   |
|              | Name: Mariana Patricia  |       |                   |

PART C: Finally, fill in a table showing the times obtained (in milliseconds) for the files mentioned above and the number of bins expected and obtained in each case. What complexity do you assign to the algorithm that calculates the minimum number of containers to use? Why?

| Repetitions: 50000 |          |      |
|--------------------|----------|------|
| TestFile           | Time(ms) | Bins |
| test00.txt         | 220      | 4    |
| test01.txt         | 20       | 1    |
| test02.txt         | 52       | 8    |
| test03.txt         | 27       | 2    |
| test04.txt         | 46       | 10   |
| test05.txt         | 160      | 14   |
| test06.txt         | 29       | 7    |
| test07.txt         | 39       | 8    |
| test08.txt         | 142      | 16   |
| test09.txt         | 243      | 33   |

The algorithm that calculates the minimum number of containers to use has an exponential complexity, specifically  $O(k^n)$ , where  $n$  is the number of objects and  $k$  is the number of bins.

As there is no known polynomial time algorithm for this problem, we implemented a backtracking algorithm, which always has exponential or factorial complexity. However, the pruning techniques I implemented, significantly optimize it.